

1. You are given:

```
X = torch.tensor([[1.0, 2.0]])
```

```
W = torch.tensor([[0.5],  
                  [0.3]])
```

```
b = torch.tensor([[0.1]])
```

Questions:

1. What are the shapes of **X**, **W**, and **b**? Write code to see shapes.
2. Will `torch.matmul(X, W)` work? Why?
3. What is the output of matrix multiplication ? (First write the code by yourself)
4. If it fails, how would you reshape the tensors correctly?
5. Manual Forward Pass of a Two-Layer Neural Network

2. You are given a two-layer neural network with the following parameters:

Input **X** is a column vector of size (2×1) .

W1 is of shape (2×2) .

b1 is of shape (2×1) .

W2 is of shape (2×1) .

b2 is of shape (1×1) .

Tasks:

- a) Write the mathematical formula for **Z1**.
- b) Apply ReLU activation to compute **A1**.
- c) Write the formula for the final output **y_hat**.
- d) If all weights are initialized to 0.1 and biases are zero, compute the final output manually for input **X** = $\begin{bmatrix} 1.0 \\ 2.0 \end{bmatrix}$.

3. Understanding Mean Squared Error (MSE)

Given:

$$y = 5.0$$

$$\hat{y} = 3.0$$

Questions:

a) Compute the squared error manually.